

**Q1. Area in terms of range**

Let the flight time be  $t$ . Resolving the initial velocity,

$$\vec{v}_0 = (v_0 \cos \theta, v_0 \sin \theta), \quad \vec{g}t = (0, -gt).$$

The cross-product magnitude of any two sides of the triangle gives (twice) its area. Choosing  $\vec{v}_0$  and  $\vec{g}t$ ,

$$2S = |\vec{v}_0 \times (\vec{g}t)| = |(v_0 \cos \theta)(-gt) - (v_0 \sin \theta)(0)| = gtv_0 \cos \theta.$$

Since  $x = v_0 \cos \theta \cdot t$  is precisely the horizontal range, hence

$$\boxed{S = \frac{1}{2}gx}.$$

**Q2. Area in angular form**

For ground-to-ground motion, the vertical displacement is zero:

$$0 = v_0 \sin \theta \cdot t - \frac{1}{2}gt^2 \implies t = \frac{2v_0 \sin \theta}{g}.$$

Substituting into the area formula of Q1,

$$S = \frac{1}{2}g(v_0 \cos \theta) \frac{2v_0 \sin \theta}{g} = \boxed{\frac{1}{2}v_0^2 \sin 2\theta}.$$

Now consider the same triangle but with the two sides  $\vec{v}_0$  and  $\vec{v}$  meeting at the origin. For ground-to-ground motion, energy conservation gives  $|\vec{v}| = |\vec{v}_0| = v_0$ . The (signed) area of the triangle is

$$S = \frac{1}{2} |\vec{v}_0| |\vec{v}| \sin \alpha = \frac{1}{2} v_0^2 \sin \alpha,$$

where  $\alpha$  is the interior angle at the origin, taken from  $\vec{v}_0$  to  $\vec{v}$ . At landing  $\vec{v} = (v_0 \cos \theta, -v_0 \sin \theta)$ , so  $\alpha = -2\theta$  (the velocity has rotated clockwise by  $2\theta$ ). Hence

$$\sin \alpha = \sin(-2\theta) = -\sin 2\theta, \quad S = -\frac{1}{2}v_0^2 \sin \alpha.$$

The two forms are therefore equivalent:  $\boxed{S = \frac{1}{2}v_0^2 \sin 2\theta = -\frac{1}{2}v_0^2 \sin \alpha}$ .

The maximum of  $S$  — and therefore of  $x$  — occurs when  $\sin 2\theta = 1$ , i.e.  $\boxed{\theta = 45^\circ}$ , recovering the familiar result through a purely geometric route.

**Q3. Non-zero launch height**

The mass is launched from  $y = h > 0$  and lands at  $y = 0$ . The flight time satisfies

$$0 = h + v_0 \sin \theta \cdot t - \frac{1}{2}gt^2,$$

giving

$$t(\theta) = \frac{v_0 \sin \theta + \sqrt{v_0^2 \sin^2 \theta + 2gh}}{g}.$$

The horizontal range is  $L(\theta) = v_0 \cos \theta \cdot t(\theta)$ .

To maximise  $L$ , set  $dL/d\theta = 0$ . Writing  $L$  explicitly and differentiating (or, more elegantly, using the implicit relation  $L \tan \theta - gL^2/(2v_0^2 \cos^2 \theta) + h = 0$  obtained by eliminating  $t$ ), one finds

$$\boxed{\sin^2 \theta_{\text{opt}} = \frac{v_0^2}{2(v_0^2 + gh)}}.$$

Equivalently,

$$\theta_{\text{opt}} = \arcsin \sqrt{\frac{v_0^2}{2(v_0^2 + gh)}}, \quad \text{or} \quad \tan \theta_{\text{opt}} = \frac{v_0}{\sqrt{v_0^2 + 2gh}}.$$

*Geometric note.* From Q1,  $S = \frac{1}{2}gL$  still holds. Energy conservation gives  $|\vec{v}|^2 = v_0^2 + 2gh$ , so the two sides meeting at the origin have *fixed* lengths  $v_0$  and  $\sqrt{v_0^2 + 2gh}$ . The area  $S = \frac{1}{2}v_0 |\vec{v}| \sin \varphi$  (where  $\varphi$  is the angle between  $\vec{v}_0$  and  $\vec{v}$ ) is maximised when  $\varphi = 90^\circ$ , i.e. when  $\vec{v}_0 \perp \vec{v}$ . Imposing orthogonality on the vector triangle yields the same optimal angle as above, without calculus.